Vision Foundation Model-Driven Multiscale Expert Tuning for Multimodal Remote Sensing Semantic Segmentation

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***Abstract*—Multimodal remote sensing semantic segmentation based on optical and digital surface model (Opt-DSM) data is pivotal for comprehensive scene interpretation. However, prevailing methodologies often lack a unified vision foundation model and encounter significant challenges in bridging modality gaps and achieving effective feature fusion. Conventional models, such as the segment anything model (SAM), exhibit inherent limitations when addressing the unique complexities of multimodal remote sensing, particularly in managing cross-modal discrepancies and intricate surface structures. In this study, we present vision foundation model-driven multiscale expert tuning (VF-MET), an innovative framework meticulously tailored for Opt-DSM semantic segmentation tasks. VF-MET incorporates an adaptive multiscale expert tuning (AMET) strategy, which substantially enhances the feature extraction capabilities of vision foundation models. This enables the robust capture of cross-scale and morphologically irregular objects, while simultaneously preserving superior generalization ability. To further address the segmentation of densely distributed and weakly correlated regions, we propose a collaborative box-point prompt mechanism (CBPM), which significantly improves spatial localization and contextual discrimination. Moreover, we introduce a two-stage mask decoder (TSMD) that facilitates efficient multimodal feature fusion and augments contextual understanding. Extensive experiments conducted on public Opt-DSM benchmark datasets unequivocally demonstrate that VF-MET achieves state-of-the-art performance. Comprehensive ablation studies further substantiate the indispensable contributions of each constituent module within the proposed architecture. The source code and datasets are publicly accessible at https://github.com/NWPUFranklee/VF-MET.git**

***Index Terms*—Feature fusion, frequency-aware modeling, remote sensing imagery, road extraction, state-space model.**

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I. Introduction

MULTIMODAL remote sensing semantic segmentation has rapidly evolved into a highly promising methodology for precise land cover analysis and comprehensive scene understanding. Recent research has convincingly demonstrated that the integration of heterogeneous data source, such as optical-SAR [1], [2], [3], [4], optical-thermal [5], [6], [7], [8], and hyperspectral-multispectral [9], [10], [11] modalities, which can markedly enhance segmentation accuracy and robustness. Within this context, the fusion of optical data and digital surface models stands out by synergistically harnessing both spectral and elevation cues through the precise alignment of optical imagery with DSMs generated from point clouds [12], [13], [14], as illustrated in Fig. 1(a). Such multimodal integration effectively mitigates the inherent deficiency of 3-D contextual information in conventional optical images, thereby significantly improving pixel-level classification in complex environments. Consequently, this approach has been widely adopted in diverse applications, including defense, land resource monitoring, and urban planning, where nuanced and accurate scene interpretation is indispensable.

Currently, multimodal remote sensing semantic segmentation methodologies can be broadly categorized into two principal paradigms, as depicted in Fig. 1(b) and (c): the pyramid feature paradigm and the segment anything model (SAM)-based nonpyramid feature paradigm. Both frameworks typically encompass three fundamental stages: extraction of optical features, extraction of cross-modal features, and fusion of multimodal representations. Among these, the pyramid paradigm [15], [16], [17], [18] has achieved widespread adoption, primarily relying on U-shaped encoder–decoder architectures to hierarchically structure multimodal information. In such frameworks, shallow network layers are responsible for capturing intricate visual details, whereas deeper layers abstract high-level semantic content. To overcome the inherent limitations of 2-D representations in optical and digital surface model (Opt-DSM) imagery and the insufficient modeling of long-range dependencies, FTransDeepLab [19] incorporates a Transformer [20] backbone into DeepLabv3+ [21], augmented with novel MFR and MFF modules. This integration markedly advances the efficacy of multimodal feature fusion. Furthermore, in response to the challenges associated with cross-modal integration and

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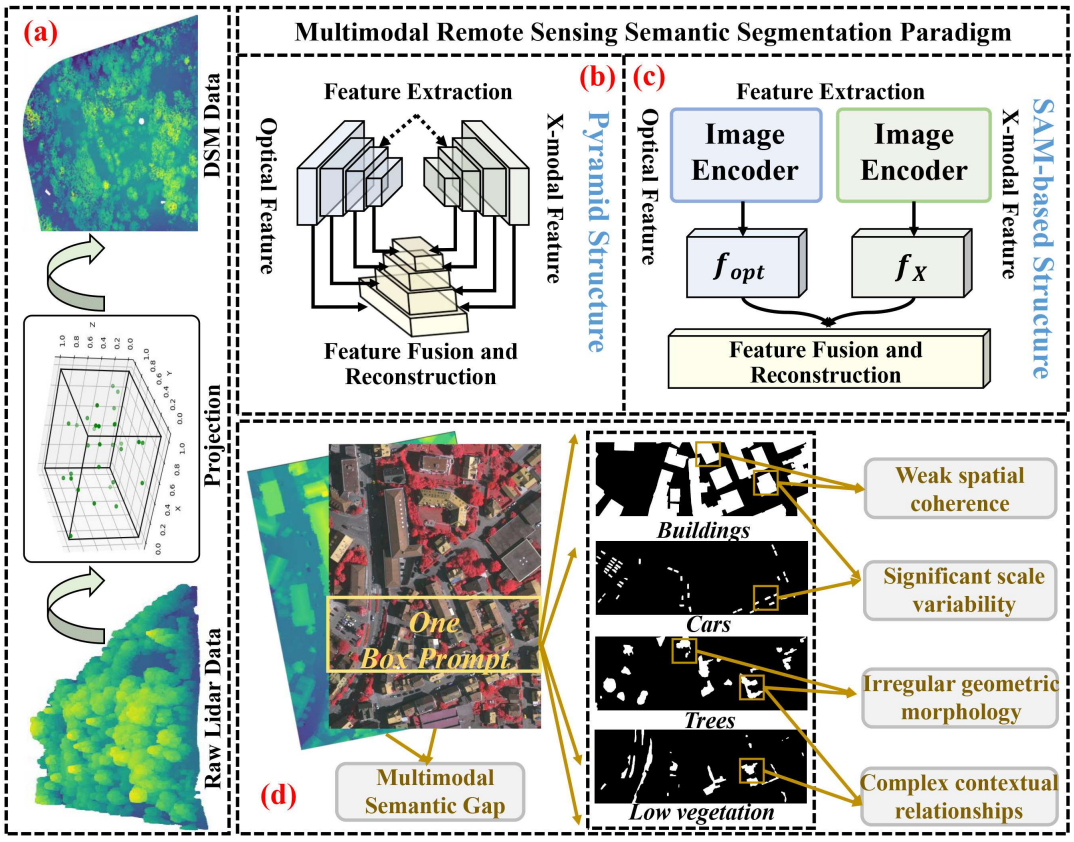


Fig. 1. Overview of the multimodal remote sensing semantic segmentation paradigm and main challenges for Opt-DSM segmentation. (a) DSM data generation via projection of raw point clouds. (b) Pyramid structure paradigm for feature extraction and fusion. (c) SAM-based structure paradigm with parallel encoders and prompt-based fusion. (d) Typical challenges in Opt-DSM segmentation: weak spatial coherence, scale variability, irregular object shapes, complex contextual relations, and multimodal semantic gap.

adaptation to heterogeneous scenes, SiMultiF [22] proposes a fusion architecture featuring a twin-network design and adaptive modal weight allocation. This enables dynamic fusion of modality-specific features and leads to notable improvements in segmentation performance across a variety of scenarios.

Although large-scale pretraining models, such as SAM [23] facilitate the extraction of rich semantic representations, the pyramid feature paradigm frequently demonstrates inferior accuracy compared to the SAM-based paradigm, particularly in delineating complex object boundaries, segmenting small-scale targets, and managing irregularly shaped ground objects. Moreover, the generalization ability of pyramid-based models is generally outperformed by the robust, prompt-driven mechanisms characteristic of SAM-based approaches. In contrast to conventional end-to-end pyramid architectures, the SAM-based paradigm [24], [25], [26], [27] has set new standards for segmentation accuracy, thereby accelerating its adoption in multimodal remote sensing applications. However, several challenges remain when deploying SAM for Opt-DSM semantic segmentation: these include restricted category recognition, diminished efficacy with DSM data, and inherent limitations associated with single-box prompt strategies. To mitigate these deficiencies, RingMo-SAM [28] introduces a CDMDecoder in conjunction with a remote sensing feature prompt encoder, thereby augmenting category segmentation precision and model adaptability. Nonetheless, the adoption of a multibox prompt scheme in RingMo-SAM brings additional challenges, particularly with respect to the optimal selection and quantity determination of prompt boxes, which increases the complexity of the segmentation process.

To ensure high segmentation accuracy in multimodal remote sensing, our methodology is fundamentally anchored in the SAM-based paradigm. Nevertheless, as illustrated in Fig. 1(d), several intrinsic challenges persist when this paradigm is applied to Opt-DSM data. First, although box prompt learning in SAM generally surpasses point prompts across various domains [29], [30], [31], [32], its effectiveness in remote sensing scenarios is considerably constrained. This limitation arises from the high object density and weak spatial coherence characteristic of such imagery: objects are often irregularly distributed and exhibit minimal spatial correlation, which frequently causes box prompts to overlap multiple targets and consequently undermines segmentation performance. Second, Opt-DSM images are marked by pronounced scale heterogeneity and morphological irregularities. For instance, the substantial size disparity between classes, such as buildings and cars, coupled with the highly irregular shapes of categories like trees, poses significant challenges. Capturing these variations in scale and form is imperative for achieving accurate boundary delineation. However, the Transformer-based architecture underpinning SAM is not inherently tailored for extracting fine-grained, scale-adaptive features necessary

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for this level of detail. Third, a pronounced semantic gap exists between optical and DSM modalities. While optical images offer rich textural information, DSM exclusively encodes elevation data. This disparity often results in ambiguous correspondences; for example, distinct objects in optical images may exhibit similar elevation values in the DSM, or conversely, the same object may appear at different elevation levels. Moreover, intricate contextual relationships both within and between modalities, such as those between trees and low vegetation, further complicate the effective fusion of multimodal features. Collectively, these challenges underscore the necessity for innovative strategies that can bridge modality gaps, accommodate scale and shape diversity, and facilitate robust multimodal feature integration in Opt-DSM semantic segmentation.

To overcome the aforementioned challenges, we propose vision foundation model-driven multiscale expert tuning (VF-MET), a vision foundation model-driven framework specifically designed for Opt-DSM multimodal remote sensing semantic segmentation. VF-MET embodies a series of pivotal innovations aimed at advancing segmentation accuracy and facilitating effective multimodal feature integration. First, we devise a collaborative box-point prompt mechanism (CBPM), which enables precise localization of segmentation regions and effectively addresses the issue of weak spatial correlation. This mechanism establishes a robust semantic foundation for subsequent multimodal feature fusion. Second, we introduce an adaptive multiscale expert tuning (AMET) strategy, wherein multiscale convolutional operations are synergistically combined with a mixture-of-experts (MoEs) architecture within the encoder. This design empowers the model to dynamically capture scale-adaptive and deformation-sensitive features, thereby achieving accurate delineation of both large-scale and morphologically irregular objects. Third, to bridge the semantic gap between optical and DSM modalities while capturing intricate contextual dependencies, we develop a two-stage mask decoder (TSMD) that leverages a dual-stage cross-attention mechanism. This decoder seamlessly integrates prompt embeddings with modality-specific features, thereby reinforcing semantic alignment, modeling complex contextual relationships, and enabling efficient multimodal fusion. The principal contributions of this work are summarized as follows.

1) We propose VF-MET, a novel interactive vision foundation framework for Opt-DSM multimodal remote sensing semantic segmentation, following the SAM paradigm. VF-MET integrates prompt-driven segmentation with multimodal fusion and achieves state-of-the-art performance on benchmark datasets, as validated by extensive comparative, ablation, and scenario-based experiments.
2) We introduce a collaborative box-point prompt embedding strategy that jointly encodes spatial priors from both optical imagery and DSM data. This approach effectively mitigates the weak spatial correlation issue in multimodal feature spaces and enables the model to establish robust intra and intermodal relationships for improved segmentation accuracy.
3) We design an encoder with AMET, incorporating a MoEs architecture and multiscale convolutional modules. This design enables the encoder to dynamically capture heterogeneous region features and complex object boundaries across scales, thereby enhancing the model's ability to segment small, irregular, and contextually ambiguous regions.
4) We develop a novel two-stage fusion strategy in the decoder, consisting of prompt-embedding fusion followed by embedding-embedding fusion. By leveraging a two-stage cross-attention mechanism, this approach achieves more effective and flexible multimodal feature interaction and integration, leading to superior segmentation performance in complex real-world scenarios.

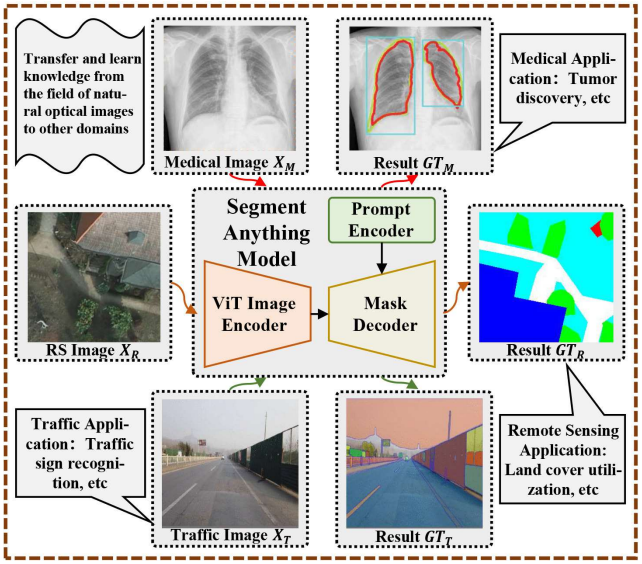


Fig. 2. Overview of the cross-domain applications of the SAM. SAM can be applied to medical imaging, remote sensing, and traffic scene analysis, enabling automatic segmentation in various fields.

The article is structured as follows. Section II provides an overview of the related works. Section III introduces the proposed VF-MET framework and its associated components. The experimental results and discussion analysis are presented in Section IV. Finally, the conclusion is drawn in Section V.

II. Related Works

A. General Vision Foundation Models

Recent advancements in vision foundation models, most notably the SAM, have established a unified and robust framework for a wide array of computer vision tasks [33], [34], [35], [36], as depicted in Fig. 2.These models have been actively investigated by researchers for a diverse range of downstream applications, demonstrating remarkable versatility in both unimodal and multimodal contexts.

1) Unimodal Applications: Within the field of medical imaging, precise segmentation is indispensable for clinical diagnosis and therapy planning. Hu et al. [37] are among the first to integrate SAM with computed tomography (CT) images, thereby establishing SAM as an effective interactive annotation tool for medical image segmentation. To address the challenges posed by modality heterogeneity and the scarcity of labels in multiparameter MRI (mpMRI) brain scans,

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Diana-Albelda et al. [38] proposed a 3-D spatial perceptual prompt encoder (3-D-SAPE), which significantly enhances feature fusion across mpMRI sequences and enables highly accurate tumor segmentation. In the domain of traffic analysis, Zhou et al. [39] introduced an adaptation layer in conjunction with a sparse hint mechanism for SAM, substantially improving crack segmentation performance in vehicle camera imagery and mitigating the generalization constraints of earlier methods. Furthermore, for urban traffic surveillance, Shokri et al. [40] developed a SAM-based algorithm that integrates DeepSORT for vehicle detection and tracking with uncalibrated, low-resolution cameras, consistently achieving superior detection, and segmentation outcomes.

2) Multimodal Applications: In the realm of multimodal medical imaging, Shi et al. [41] proposed a multimodal fusion adapter to fine-tune SAM, thereby enabling accurate glioma segmentation across a wide spectrum of imaging modalities. Likewise, Gu et al. [42] devised a three-stage architecture based on SAM, employing dedicated adapters and contrastive learning to effectively integrate CT images with high-dimensional radiomics features, thus overcoming the inherent limitations of single-modality analyses. In intelligent transportation, Li et al. [43] introduced the FusionSAM framework, which combines LSTG and FMP modules to achieve efficient multimodal fusion and segmentation. FusionSAM has been shown to outperform both SAM and SAM2 in autonomous driving scenarios. While these innovative approaches offer valuable perspectives for adapting SAM to specialized domains, their direct applicability to high-resolution, fine-grained remote sensing data remains limited. This is primarily due to domain-specific challenges, such as pronounced scale variation, spatial heterogeneity, and complex multimodal semantic gaps.

B. Remote Sensing Vision Foundation Models

1) Unimodal Applications: Vision foundation models have been increasingly leveraged in a variety of remote sensing tasks. For multispectral crop segmentation, Song et al. [44] proposed the MAF-SAM approach, which incorporates multistage adaptive fine-tuning utilizing prefix adapters and low-rank adaptation (LoRA), thereby substantially enhancing segmentation performance as well as both temporal and spatial transferability. In the realm of few-shot semantic segmentation, Ao et al. [45] introduced SEMPNet, a framework that leverages SAM-assisted mask generation in tandem with cross-image location attention and mask parsing modules, ultimately achieving markedly superior results compared to existing methodologies. For the specific task of road extraction from high-resolution remote sensing imagery, Feng et al. [46] presented Road-SAM, which integrates adapter variants, an EVP mechanism, and frequency adapter fine-tuning, collectively resulting in notable improvements in segmentation accuracy.

2) Multimodal Applications: Concurrently, substantial efforts have been devoted to extending vision foundation models to multimodal remote sensing scenarios. For instance, in Opt-SAR image matching, SFA-Net [47] has been proposed to mitigate nonlinear radiometric distortions and geometric discrepancies by means of cross-modal feature representation and difference matching. Nevertheless, its efficacy diminishes in texture-sparse environments, and the integration of SAM further escalates computational complexity. In the context of infrared object detection, Zhang et al. [48] introduced the MIRSAM framework, which optimizes SAM with a CDAM module and incorporates the CLIP [49] text encoder, thereby enhancing both accuracy and robustness. Wang et al. [50] proposed an auto-prompting framework that adaptively generates both box and point prompts from class activation maps to guide SAM for weakly supervised landslide extraction. For open-vocabulary segmentation, Liu et al. [51] devised a method that integrates large language models (LLMs) with SAM, empowering the segmentation of previously unseen categories. While this strategy improves generalization capacity, it tends to underperform relative to task-specific models and introduces significant complexity. Ma et al. [24] proposed MFNet, a unified multimodal fine-tuning framework based on SAM, which achieves state-of-the-art performance on several benchmark remote sensing segmentation datasets. Despite these advancements, current methodologies remain insufficient to address the distinct challenges inherent in Opt-DSM semantic segmentation, such as modality gaps, scale heterogeneity, and contextual dependencies.

III. Methodology

In this section, we present the detailed architecture of VF-MET, a novel foundational model for promptable Opt-DSM multimodal remote sensing semantic segmentation. As depicted in Fig. 3, VF-MET consists of three core modules.

1) Efficient and AMET strategy, implemented as a ViT backbone enhanced with MoE multiscale convolutional LoRA (MoE-MSConvLora). This design preserves the pretrained knowledge of SAM while enabling dynamic adaptation to the domain-specific characteristics of multimodal remote sensing data [Fig. 3(a)].
2) CBPM, which integrates both box and point prompts to facilitate information association across modalities and improve the precise localization of segmentation regions [Fig. 3(b)].
3) TSMD, derived from the original SAM decoding framework, which enables effective multimodal feature fusion and accurate mask prediction through a two-stage cross-attention mechanism [Fig. 3(c)]. These components collectively endow VF-MET with strong generalization ability and precise segmentation performance.

A. Overall Framework

In the context of multimodal semantic segmentation, we denote the dataset as $\mathcal{D} = \{\boldsymbol{I}_{\text{opt}}, \boldsymbol{I}_{\text{dsm}}, \mathbf{GT}_{(i,j)}\}$, where $\boldsymbol{I}_{\text{opt}}$ is the optical image, $\boldsymbol{I}_{\text{dsm}}$ represents the normalized DSM image, and $\mathbf{GT}_{(i,j)}$ corresponds to the ground truth label at pixel location (i, j). The optical image is formulated as $\boldsymbol{I}_{\text{opt}} \in \mathbb{R}^{3\times H\times W}$, while the DSM image is given by $\boldsymbol{I}_{\text{dsm}} \in \mathbb{R}^{1\times H\times W}$. Both modalities are first input into a fine-tuned SAM encoder, denoted as $\mathcal{F}_{\text{IE}}$, which is constructed upon a vision transformer (ViT-b) [52] backbone with shared weights. To ensure compatibility, the single-channel DSM image is expanded to three channels via channel-wise duplication. The encoder then extracts nonpyramidal multimodal features, resulting in feature maps

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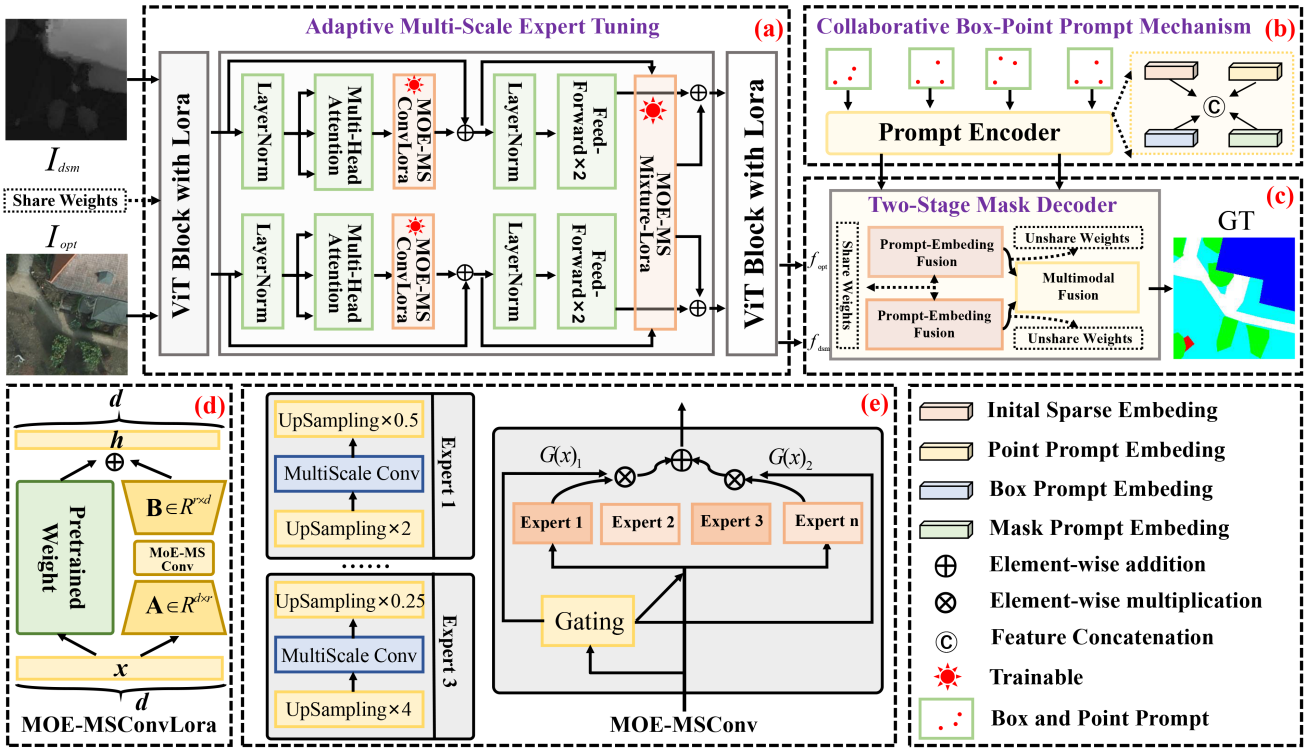


Fig. 3. Overview of the proposed VF-MET framework for multimodal remote sensing semantic segmentation. (a) AMET employs a ViT backbone with shared weights and MoE-MSConvLora for efficient feature extraction from both optical and DSM images. (b) CBPM generates and encodes both box and point prompts to enhance spatial localization. (c) TSMD integrates prompt embeddings and multimodal features for accurate segmentation prediction. (d) Illustration of the MoE-MSConvLora design for adaptive feature extraction. (e) Structure of the MoE-MSConv module, which utilizes multiple experts with gating for multiscale feature representation. The legend explains key symbols used throughout the framework.

$\boldsymbol{f}_{\text{opt}} \in \mathbb{R}^{256\times64\times64}$ and $\boldsymbol{f}_{\text{dsm}} \in \mathbb{R}^{256\times64\times64}$ for the optical and DSM branches, respectively. The use of shared parameters facilitates the extraction of modality-invariant representations while reducing computational overhead

$$\boldsymbol{f}_k = \mathcal{F}_{\text{IE}}(\boldsymbol{I}_k), \quad k \in \{\text{opt}, \text{dsm}\}. \tag{1}$$

To address the weak spatial correlation often encountered in remote sensing segmentation, we utilize both box and point prompts, denoted as $\mathcal{P}_{\text{box}}$ and $\mathcal{P}_{\text{point}}$, respectively. These prompts are derived from the ground truth $\mathbf{GT}_{(i,j)}$ and subsequently input into the prompt encoder $\mathcal{F}_{\text{PE}}$. This stage yields a sparse embedding $\boldsymbol{f}_{\text{se}}$ and a dense embedding $\boldsymbol{f}_{\text{de}}$ for subsequent multimodal feature fusion and precise localization

$$\boldsymbol{f}_{\text{se}}, \boldsymbol{f}_{\text{de}} = \mathcal{F}_{\text{PE}}(\mathcal{P}_{\text{box}}, \mathcal{P}_{\text{point}}). \tag{2}$$

Furthermore, we adopt a two-stage decoding strategy in which prompt-feature (PE) fusion precedes feature-level fusion. All prompt embeddings and multimodal features are fed into the TSMD $\mathcal{F}_{\text{TSMD}}$, which is designed to capture contextual dependencies and enable efficient multimodal integration. The decoder reconstructs the fused representations to produce the final predicted segmentation map $\boldsymbol{f}_{\text{res}}$

$$\boldsymbol{f}_{\text{res}} = \mathcal{F}_{\text{TSMD}}(\boldsymbol{f}_{\text{opt}}, \boldsymbol{f}_{\text{dsm}}, \boldsymbol{f}_{\text{se}}, \boldsymbol{f}_{\text{de}}) \tag{3}$$

where $\boldsymbol{f}_{\text{res}}$ denotes the final segmentation output, effectively integrating multimodal cues and prompt information to achieve highly accurate semantic segmentation. The complete workflow of VF-MET is depicted in Algorithm 1.

Algorithm 1 Overall Framework of VF-MET for Multimodal Remote Sensing Semantic Segmentation

Require: Dataset $\mathcal{D} = \{\boldsymbol{I}_{\text{opt}}, \boldsymbol{I}_{\text{dsm}}, \boldsymbol{GT}\}$, number of training iterations N

Ensure: Predicted segmentation maps $\boldsymbol{f}_{\text{res}}$

for each iteration $t = 1$ to N **do**
  for each sample $(\boldsymbol{I}_{\text{opt}}, \boldsymbol{I}_{\text{dsm}}, \boldsymbol{GT})$ in minibatch **do**
  // **Feature Extraction**
  $\boldsymbol{f}_{\text{opt}} \leftarrow \mathcal{F}_{\text{IE}}(\boldsymbol{I}_{\text{opt}})$
  $\boldsymbol{f}_{\text{dsm}} \leftarrow \mathcal{F}_{\text{IE}}(\boldsymbol{I}_{\text{dsm}})$
  // **Prompt Generation**
  Generate box prompt $\mathcal{P}_{\text{box}}$ and point prompt $\mathcal{P}_{\text{point}}$ from $\boldsymbol{GT}$
  // **Prompt Encoding**
  $(\boldsymbol{f}_{\text{se}}, \boldsymbol{f}_{\text{de}}) \leftarrow \mathcal{F}_{\text{PE}}(\mathcal{P}_{\text{box}}, \mathcal{P}_{\text{point}})$
  // **Mask Decoding and Fusion**
  $\boldsymbol{f}_{\text{res}} \leftarrow \mathcal{F}_{\text{TSMD}}(\boldsymbol{f}_{\text{opt}}, \boldsymbol{f}_{\text{dsm}}, \boldsymbol{f}_{\text{se}}, \boldsymbol{f}_{\text{de}})$
  // **Loss Computation and Backpropagation**
  Compute loss $L = \text{Loss}(\boldsymbol{f}_{\text{res}}, \boldsymbol{GT})$
  Update model parameters via backpropagation
  end for
end for

B. Collaborative Box-Point Prompt Mechanism

To tackle the challenge of precise segmentation region localization in multimodal remote sensing imagery, we introduce a CBPM, which jointly generates and encodes both box and

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point prompts for each target object, as illustrated in Fig. 3(b). Concretely, all label images $\mathbf{GT}^*_{(i,j)}$ are initially converted to one-hot encoded masks, where $\mathbf{GT}^*_{(i,j)} = 1$ denotes foreground pixels and $\mathbf{GT}^*_{(i,j)} = 0$ denotes background. For each object, the bounding box prompt $\mathcal{P}_{\text{box}} = [x_0, y_0, x_1, y_1]$ is determined by extracting the extremal coordinates of its foreground region

$$\mathcal{M} = \left\{(i,j) \mid \mathbf{GT}^*_{(i,j)} = 1\right\} \tag{4}$$

$$x_0 = \min\{\, i \mid (i,j) \in \mathcal{M}\,\}, \quad y_0 = \min\{\, j \mid (i,j) \in \mathcal{M}\,\} \tag{5}$$

$$x_1 = \max\{\, i \mid (i,j) \in \mathcal{M}\,\}, \quad y_1 = \max\{\, j \mid (i,j) \in \mathcal{M}\,\}. \tag{6}$$

To further augment spatial guidance, we construct a point prompt set $\mathcal{P}_{\text{point}} = [\mathcal{P}_0, \mathcal{P}_1, \ldots, \mathcal{P}_n]$. The primary prompt $\mathcal{P}_0 = (\mathcal{C}_x, \mathcal{C}_y)$ is defined as the centroid of the foreground region, computed via image moments

$$\mathcal{T}_{pq} = \sum_{i=1}^{H}\sum_{j=1}^{W} i^p j^q \mathbf{GT}^*_{(i,j)} \tag{7}$$

$$\mathcal{C}_x = \frac{\mathcal{T}_{10}}{\mathcal{T}_{00}}, \quad \mathcal{C}_y = \frac{\mathcal{T}_{01}}{\mathcal{T}_{00}}. \tag{8}$$

Additional points are sampled uniformly within the foreground mask to provide richer spatial cues. Both box and point prompts are subsequently embedded by the prompt encoder $\mathcal{F}_{\text{EB}}$ and concatenated to form the sparse embedding $\boldsymbol{f}_{\text{se}}$

$$\boldsymbol{f}_{\text{box}} = \mathcal{F}_{\text{EB}}\left(\mathcal{P}_{\text{box}}\right) \tag{9}$$

$$\boldsymbol{f}_{\text{point}} = \mathcal{F}_{\text{EB}}\left(\mathcal{P}_{\text{point}}\right) \tag{10}$$

$$\boldsymbol{f}_{\text{se}} = \mathcal{F}_{\text{Cat}}\left(\boldsymbol{f}_{\text{box}} \boldsymbol{f}_{\text{point}}\right). \tag{11}$$

As mask prompts are not leveraged in this framework, the dense embedding $\boldsymbol{f}_{\text{de}}$ is initialized as a set of learnable parameters. This collaborative strategy effectively improves localization accuracy and provides robust semantic guidance for downstream multimodal fusion.

C. Adaptive Multiscale Expert Tuning

To overcome the limitations of the original SAM encoder, which lacks domain-specific semantic understanding of remote sensing optical and DSM imagery and fails to adequately capture multiscale and irregular object structures, we propose an AMET strategy, as depicted in Fig. 3(a). Specifically, we introduce a MoE-MSConvLora module, denoted as $\mathcal{F}_{\text{M_LoRA}}$, for targeted fine-tuning of both the multihead self-attention module $\mathcal{F}_{\text{L_MHA}}$ and the mixture feedforward neural network $\mathcal{F}_{\text{L_FFN}}$ within each encoder block. In our architecture, the primary parameters are shared across modalities, while the fine-tuning modules $\mathcal{F}_{\text{M_LoRA}}$ are maintained separately for the optical and DSM branches. This design allows the model to effectively leverage both shared and modality-specific knowledge, enhancing the extraction of multiscale and diverse features from multimodal inputs. The computations within each ViT block incorporating LoRA are as follows:

$$\mathcal{Z}_{\text{in},\mathcal{K}} = \mathcal{F}_{\text{L_MHA}}\left(\mathcal{F}_{\text{Norm}}\left(\mathcal{X}_{\text{in},\mathcal{K}}\right)\right) + \mathcal{X}_{\text{in},\mathcal{K}} \tag{12}$$

$$\mathcal{X}_{\text{out},\mathcal{K}} = \mathcal{F}_{\text{L_FFN}}\left(\mathcal{F}_{\text{Norm}}\left(\mathcal{Z}_{\text{in},\mathcal{K}}\right)\right) + \mathcal{Z}_{\text{in},\mathcal{K}} \tag{13}$$

where $\mathcal{F}_{\text{Norm}}$ denotes LayerNorm, $\mathcal{X}_{\text{in},\mathcal{K}}$ is the input feature for modality $\mathcal{K}$ (either optical or DSM), $\mathcal{Z}_{\text{in},\mathcal{K}}$ is the intermediate representation, and $\mathcal{X}_{\text{out},\mathcal{K}}$ is the output of the block. For the fine-tuning of the multihead self-attention mechanism, the operations are as follows:

$$\mathcal{S}^{\mathcal{L}}_{\mathcal{K}} = \mathcal{F}_{\text{M_LoRA}}\left(\mathcal{X}_{\text{in},\mathcal{K}}\right), \quad \mathcal{S} \in \{Q, K, V\} \tag{14}$$

$$\mathcal{F}^{\mathcal{L}}_{\text{L_MHA}} = \left(\frac{Q^{\mathcal{L}}_{\mathcal{K}} K^{\mathcal{L}\top}_{\mathcal{K}}}{\sqrt{d}}\right) V^{\mathcal{L}}_{\mathcal{K}} \tag{15}$$

where $\mathcal{S}^{\mathcal{L}}_{\mathcal{K}}$ denotes the fine-tuned query, key, and value matrices for modality $\mathcal{K}$, and d represents the feature dimension. The mixture feedforward neural network is computed as

$$\mathcal{Z}_{\text{in},\mathcal{K}} = \mathcal{F}_{\text{M_LoRA}}\left(\text{GeLU}\left(\mathcal{F}_{\text{M_LoRA}}\left(\mathcal{X}_{\text{in},\mathcal{K}}\right)\right)\right) \tag{16}$$

where GeLU$(\cdot)$ denotes the Gaussian error linear unit activation. To further facilitate cross-modal interaction, the outputs for the optical and DSM modalities are adaptively fused using learnable weights λ_{opt} and λ_{dsm} (both initialized to 1)

$$\mathcal{X}_{\text{out,opt}} = \lambda_{\text{opt}} \mathcal{Z}_{\text{in,opt}} + \left(1 - \lambda_{\text{opt}}\right) \mathcal{Z}_{\text{in,dsm}} \tag{17}$$

$$\mathcal{X}_{\text{out,dsm}} = \lambda_{\text{dsm}} \mathcal{Z}_{\text{in,dsm}} + \left(1 - \lambda_{\text{dsm}}\right) \mathcal{Z}_{\text{in,opt}} \tag{18}$$

where λ_{opt} and λ_{dsm} modulate the information flow between the two modalities. This adaptive expert tuning mechanism empowers the encoder to capture fine-grained, multiscale, and modality-sensitive representations, thus significantly improving segmentation performance in multimodal scenarios.

D. MoE-MSConvLora

To address the dual challenges of efficient fine-tuning and robust multiscale modeling in multimodal remote sensing, we propose the MoE-MSConvLora module, which integrates LoRA [53] with a MoEs multiscale convolutional design, as illustrated in Fig. 3(d). The conventional LoRA framework utilizes a pair of learnable low-rank matrices to approximate weight updates, which greatly reduces the number of trainable parameters while maintaining (or even enhancing) model performance. Given a pretrained weight matrix $\mathcal{W}_{\text{pre}} \in \mathbb{R}^{d\times k}$, LoRA introduces a trainable low-rank component $\Delta\mathcal{W}_{\text{LoRA}}$, yielding the fine-tuned weight as follows:

$$\mathcal{W}_{\text{finetune}} = \mathcal{W}_{\text{pre}} + \Delta\mathcal{W}_{\text{LoRA}} \tag{19}$$

$$\Delta\mathcal{W}_{\text{LoRA}} = \mathcal{W}_B \mathcal{W}_A \tag{20}$$

where $\mathcal{W}_A \in \mathbb{R}^{r\times k}$ is the down-projection matrix, $\mathcal{W}_B \in \mathbb{R}^{d\times r}$ is the up-projection matrix, and $r \ll \min(d,k)$ represents the adaptation rank. To further enhance the model's ability to capture irregular structures and multiscale features, we incorporate an MoE-MSConv between $\mathcal{W}_A$ and $\mathcal{W}_B$. The input is first projected into a low-rank space

$$\mathcal{Z}_{\text{in},\mathcal{K}} = \mathcal{W}_A \mathcal{X}_{\text{in},\mathcal{K}} \tag{21}$$

and the pretrained feature is computed as $h = \mathcal{W}_{\text{pre}} \mathcal{X}_{\text{in},\mathcal{K}}$. The output after expert fusion is given by

$$\mathcal{X}_{\text{out},\mathcal{K}} = \mathcal{W}_B \left(\sum_{i=1}^{n}\left[\mathcal{G}\left(\mathcal{Z}_{\text{in},\mathcal{K}}\right)_i \, \mathcal{E}_i\left(\mathcal{Z}_{\text{in},\mathcal{K}}\right)\right]\right) + h \tag{22}$$

where $\mathcal{G}(\cdot)_i$ is the gating activation for the ith expert, and $\mathcal{E}_i(\cdot)$ denotes the ith expert system, each specializing in different multiscale spatial representations. Each expert comprises

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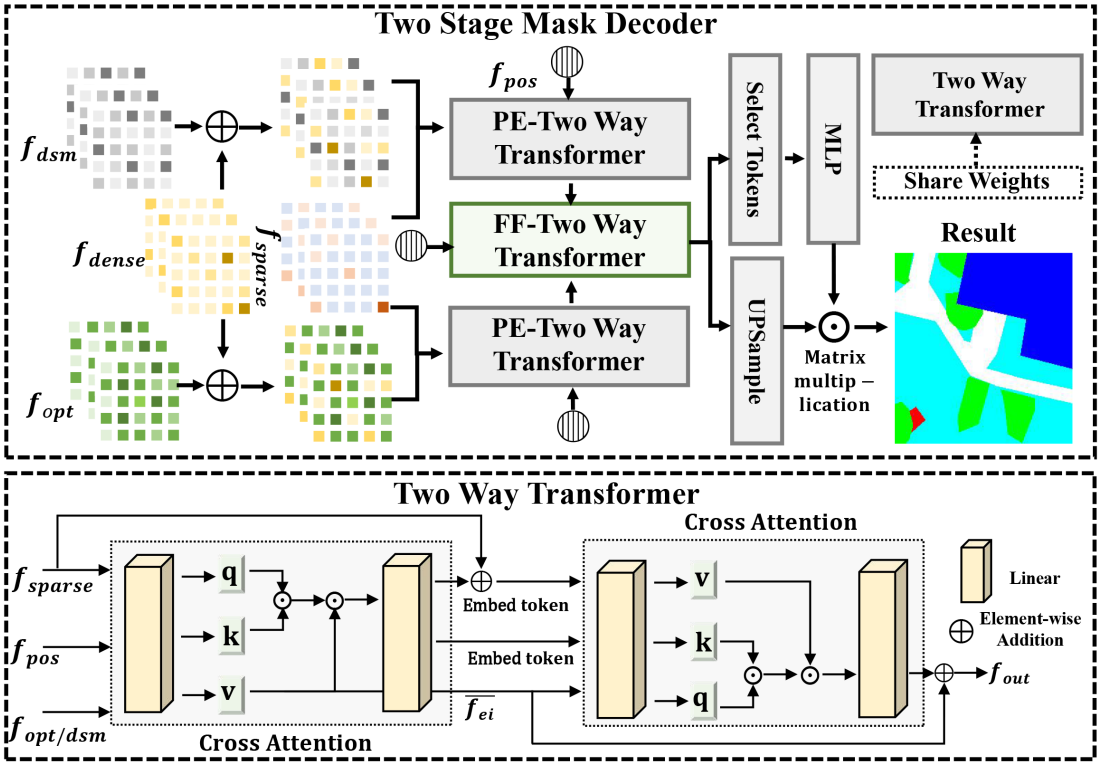


Fig. 4. Illustration of the TSMD. (Upper) Sequential fusion of optical, DSM, and prompt features via prompt-feature and feature-feature two-way transformers, followed by upsampling and final mask prediction. (Lower) Two-way transformer, which leverages bidirectional cross-attention between prompt embeddings and modality features for enhanced multimodal contextual integration.

upsampling, downsampling, and multiscale convolution operations as follows:

$$\mathcal{Z}_{\text{in},\mathcal{K}}^{s_i} = \mathcal{F}_{\text{up}\times s_i}\left(\mathcal{X}_{\text{in},\mathcal{K}}\right) \tag{23}$$

$$\mathcal{S}_{\text{in},\mathcal{K}}^{s_i} = \mathcal{F}_{\text{MSConv}}\left(\mathcal{Z}_{\text{in},\mathcal{K}}^{s_i}\right) \tag{24}$$

$$\mathcal{X}_{\text{out},\mathcal{K}}^{s_i} = \mathcal{F}_{\text{down}\times\frac{1}{s_i}}\left(\mathcal{S}_{\text{in},\mathcal{K}}^{s_i}\right) \tag{25}$$

where s_i denotes the upsampling scale factor for the ith expert. The multiscale convolutional feature for each scale is constructed as follows:

$$\mathcal{S}_{\text{in},\mathcal{K}}^{s_i\times 1} = \text{GeLU}\left(\mathcal{F}_{3\times 3}\left(\mathcal{Z}_{\text{in},\mathcal{K}}\right)\right) \tag{26}$$

$$\mathcal{S}_{\text{in},\mathcal{K}}^{s_i\times 2} = \text{GeLU}\left(\mathcal{F}_{5\times 5}\left(\mathcal{Z}_{\text{in},\mathcal{K}}\right)\right) \tag{27}$$

$$\mathcal{S}_{\text{in},\mathcal{K}}^{s_i\times 3} = \text{GeLU}\left(\mathcal{F}_{7\times 7}\left(\mathcal{Z}_{\text{in},\mathcal{K}}\right)\right) \tag{28}$$

$$\mathcal{S}_{\text{in},\mathcal{K}}^{s_i} = \text{GeLU}\left(\mathcal{F}_{1\times 1}\left(\mathcal{F}_{\text{Cat}}\left[\mathcal{S}_{\text{in},\mathcal{K}}^{s_i\times 1}, \mathcal{S}_{\text{in},\mathcal{K}}^{s_i\times 2}, \mathcal{S}_{\text{in},\mathcal{K}}^{s_i\times 3}\right]\right)\right) \tag{29}$$

where $\mathcal{F}_{c\times c}$ denotes a convolution with kernel size c, and $\mathcal{F}_{\text{Cat}}$ indicates channel-wise concatenation. The MoE-MSConvLora module thereby enables adaptive, expert-driven, and multiscale feature extraction, greatly enhancing the model's ability to represent and segment complex, irregular objects in multimodal remote sensing imagery.

E. Two-Stage Mask Decoder

To address the dual challenges of efficient multimodal feature integration and semantic gap bridging between optical and DSM modalities, we introduce the TSMD. This decoder is specifically engineered to exploit the complementary properties of optical images, characterized by rich textures and visual details, and DSM data, which provide precise boundary and elevation information, while simultaneously mitigating inherent semantic discrepancies. The proposed decoder adopts a sequential two-stage fusion strategy, as illustrated in Fig. 4. In the first stage, a shared-parameter PE two-way transformer ($\mathcal{F}_{\text{PF}}$) is employed to integrate prompt embeddings with modality-specific features. This enhances intramodal contextual representation and leverages the prompt as a semantic bridge to reduce modality gaps. In the second stage, a feature-feature (FF) two-way transformer ($\mathcal{F}_{\text{FF}}$, without parameter sharing) is utilized to achieve deep fusion and contextual correlation between the two modalities. The overall computation of the decoder $\mathcal{F}_{\text{TSMD}}$ is as follows:

$$\boldsymbol{f}_{\mathcal{K}} = \boldsymbol{f}_{\mathcal{K}} + \boldsymbol{f}_{\text{de}} \tag{30}$$

$$\boldsymbol{f}_{\text{out}_\mathcal{K}} = \mathcal{F}_{\text{PF}}\left(\boldsymbol{f}_{\text{se}}, \boldsymbol{f}_{\text{pos}}, \boldsymbol{f}_{\mathcal{K}}\right) \tag{31}$$

$$\boldsymbol{f}_{\text{out}} = \mathcal{F}_{\text{FF}}\left(\boldsymbol{f}_{\text{out_dsm}}, \boldsymbol{f}_{\text{pos}}, \boldsymbol{f}_{\text{out_opt}}\right) \tag{32}$$

where $\boldsymbol{f}_{\text{out}_\mathcal{K}}$ denotes the fused embedding of the prompt and modality $\mathcal{K}$, $\boldsymbol{f}_{\text{out}}$ is the final multimodal fused embedding, and $\boldsymbol{f}_{\text{pos}}$ denotes positional encoding. Within the PE two-way transformer, the operations are defined as follows:

$$\mathcal{F}_{\text{MHA}}\left(Q, K, V\right) = \left(\frac{QK^{\top}}{\sqrt{d}}\right)V \tag{33}$$

$$Q_{\text{out}} = \mathcal{F}_{\text{FFN}}\left(\mathcal{F}_{\text{MHA}}\left(\boldsymbol{f}_{\text{se}}, \boldsymbol{f}_{\mathcal{K}}, \boldsymbol{f}_{\mathcal{K}}\right) + \boldsymbol{f}_{\text{se}}\right) \tag{34}$$

$$Q_1 = Q_{\text{out}} + \boldsymbol{f}_{\text{se}}, \quad K_1 = \boldsymbol{f}_{\mathcal{K}} + \boldsymbol{f}_{\text{pos}} \tag{35}$$

$$K_{\text{out}} = \mathcal{F}_{\text{MHA}}\left(Q = K_1, K = Q_1, V = Q_{\text{out}}\right) + \boldsymbol{f}_{\mathcal{K}} \tag{36}$$

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where $\mathcal{F}_{\mathrm{MHA}}$ denotes the multihead attention operation. Similarly, the FF two-way transformer is as follows:

$$Q_{\mathrm{out}} = \mathcal{F}_{\mathrm{FFN}}\left(\mathcal{F}_{\mathrm{MHA}}\left(f_{\mathrm{dsm}}, f_{\mathrm{opt}}, f_{\mathrm{opt}}\right) + f_{\mathrm{dsm}}\right) \tag{37}$$

$$R_1 = Q_{\mathrm{out}} + f_{\mathrm{dsm}} \tag{38}$$

$$K_1 = f_{\mathrm{opt}} + f_{\mathrm{pos}} \tag{39}$$

$$K_{\mathrm{out}} = \mathcal{F}_{\mathrm{MHA}}\left(Q = K_1, K = R_1, V = Q_{\mathrm{out}}\right) + f_{\mathrm{opt}}. \tag{40}$$

After multimodal feature integration, subsequent decoding operations strictly adhere to the original design philosophy of SAM, thereby preserving its established strengths and innovative advantages.

F. Loss Function

To address the pronounced foreground–background pixel imbalance inherent in multimodal remote sensing image segmentation, we employ DiceCELoss as the principal optimization objective. This composite loss integrates dice loss and cross-entropy loss in a weighted fashion, thereby achieving a balance between region-level overlap and pixel-wise classification accuracy. The overall loss function is formulated as

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{Dice}} + \mathcal{L}_{\mathrm{CE}} \tag{41}$$

where $\mathcal{L}_{\mathrm{Dice}}$ and $\mathcal{L}_{\mathrm{CE}}$ denote the dice loss and cross-entropy loss components, respectively. Dice loss, derived from the dice coefficient, is particularly suitable for handling class imbalance in segmentation tasks. Let p_i represents the predicted probability for pixel i, y_i denotes the ground truth label, and N be the total number of pixels. The dice loss is defined as

$$\mathcal{L}_{\mathrm{Dice}} = 1 - \frac{2\sum_{i=1}^{N} y_i p_i}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} p_i}. \tag{42}$$

The cross-entropy loss measures the divergence between the predicted and actual distributions

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log\left(p_i\right) + \left(1 - y_i\right)\log\left(1 - p_i\right)\right]. \tag{43}$$

While dice loss prioritizes region-level overlap and demonstrates robustness to class imbalance, cross-entropy loss enhances pixel-wise discrimination but may be adversely affected by background dominance in highly imbalanced settings. Their combination thus ensures not only precise boundary delineation but also stable and discriminative training throughout the segmentation process.

IV. Experiments and Analysis

A. Dataset Description

In this study, we utilize two widely adopted benchmark datasets for multimodal remote sensing semantic segmentation: Potsdam and Vaihingen. Both datasets provide high-resolution true orthophotos and normalized digital surface models (nDSMs), enabling comprehensive evaluation of segmentation performance across diverse urban environments. The **Potsdam dataset** consists of 24 orthophotos with a spatial resolution of 6000 × 6000 pixels and a ground sampling distance (GSD) of 5 cm. Each image contains multiple spectral bands—infrared, red, green, and blue (IRRGB), as well as an nDSM. In our experiments, only the red, green, and blue channels are used. The dataset covers a mix of urban and non-urban areas with significant class distribution variability. For evaluation, 18 tiles (6_10, 7_10, 2_12, 3_11, 2_10, 7_8, 5_10, 3_12, 5_12, 7_11, 7_9, 6_9, 7_7, 4_12, 6_8, 6_12, 6_7, and 4_11) are used for training, and 6 tiles (2_11, 3_10, 4_10, 5_11, 6_11, and 7_12) for testing. Manual annotations are provided for six foreground classes: building, tree, low vegetation, car, impervious surface, and clutter. The **Vaihingen dataset** contains 16 orthophotos with an average tile size of 2500 × 2000 pixels and a GSD of 9 cm. Each image includes three spectral channels, near-infrared (NIR), red, and green (NIRRG), along with an nDSM. The dataset is split into 12 tiles (tiles 1, 3, 23, 26, 7, 11, 13, 28, 17, 32, 34, and 37) for training and 4 tiles (tiles 5, 21, 15, and 30) for testing. Five foreground classes are annotated, consistent with those in Potsdam except for the clutter category, which is not present. Both datasets present considerable challenges due to high spatial resolution, diverse object scales, and significant intraclass variability, providing a robust testbed for evaluating multimodal segmentation methods.

B. Implementation Details

All experiments are conducted on a workstation equipped with an Intel[1] Xeon[1] Gold 6133 CPU @ 2.50 GHz, 24 GB RAM, an NVIDIA GeForce RTX 3090 GPU, and Ubuntu 18.04 operating system. The implementation of all models is based on PyTorch 2.0.0. For model optimization, we employ the AdamW optimizer, which is effective in mitigating overfitting and ensuring stable gradient updates. Following the hyperparameter settings of SAM and further refined through grid search, VF-MET is trained for 50 epochs with an initial learning rate of 0.0001 and a weight decay of 0.01. Due to GPU constraints, the batch size during training is set to 1. Unless otherwise specified, all experimental settings and training protocols remain consistent across comparative baselines to ensure fair and rigorous evaluation.

C. Evaluation Metrics

The performance of multimodal semantic segmentation is assessed using standard metrics: overall accuracy (OA), mean $F1$-score (mF1), and mean intersection over union (mIoU).

1) *Overall Accuracy:*

$$\mathrm{OA} = \frac{1}{P}\sum_{p=1}^{P} L\left(\hat{y}_p = y_p\right) \tag{44}$$

where P denotes the total number of pixels, $\hat{y}_p$ is the predicted label, y_p is the ground truth, and $L(\cdot)$ is the indicator function that equals 1 if its argument is true and 0 otherwise.

2) *Mean F1-Score:*

$$F1_c = \frac{2 \times \mathrm{Precision}_c \times \mathrm{Recall}_c}{\mathrm{Precision}_c + \mathrm{Recall}_c} \tag{45}$$

$$\mathrm{mF1} = \frac{1}{K}\sum_{c=1}^{K} F1_c \tag{46}$$

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TABLE I

EXPERIMENTAL RESULTS ON THE VAIHINGEN DATASET. WE PRESENT THE OA OF FIVE FOREGROUND CLASSES AND THREE OVERALL PERFORMANCE METRICS. THE RED VALUES ARE THE BEST MODEL PERFORMANCE

Type	Method	OA(%)						mF1(%)	mIoU(%)
		Bui.	Tre.	Low.	Car	Imp.	Total		
Single Modal	ABCNet	94.10	90.81	78.53	64.12	89.70	89.25	85.34	75.20
	MAResU-Net	94.84	89.99	79.09	85.89	92.19	90.17	88.54	79.89
	UNetFormer	96.23	91.85	79.95	86.99	91.85	91.17	89.85	81.97
Multi-Modal	TransUNet	96.48	92.77	76.14	69.56	91.66	90.96	87.34	78.26
	CMFNet	97.17	90.82	80.37	85.47	92.36	91.40	89.48	81.44
	vFuseNet	95.92	91.36	77.64	76.06	91.85	90.49	87.89	78.92
	FuseNet	96.28	90.28	78.98	81.37	91.66	90.51	87.71	78.71
	ESANet	95.69	90.50	77.16	85.46	91.39	90.61	88.18	79.42
	SA-GATE	94.84	92.56	81.29	87.79	91.69	91.10	89.81	81.27
	CMGFNet	97.75	91.60	80.03	87.28	92.35	91.72	90.00	82.26
	MIEFNet	98.35	93.17	69.16	88.30	90.69	90.30	87.51	78.42
	MultiSenseSeg	96.79	93.20	77.98	90.99	91.82	91.47	89.94	82.15
	MFNet	97.66	90.17	82.79	87.65	92.78	91.21	91.07	84.14
	MFTransNet	96.41	91.48	80.09	86.52	92.11	91.22	89.62	81.61
	FTransUNet	98.20	91.94	81.49	91.27	93.01	92.40	91.21	84.23
	CFCGMet	98.48	92.72	81.68	91.66	93.36	92.78	91.75	84.89
	VF-MET(Ours)	96.57	96.14	92.46	97.81	96.94	95.88	96.24	87.12

where K is the number of classes, and $F1_c$ is the $F1$-score for class c, computed from its precision and recall.

3) *Mean Intersection Over Union:*

$$\mathrm{IoU}_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c} \tag{47}$$

$$\mathrm{mIoU} = \frac{1}{K}\sum_{c=1}^{K} \mathrm{IoU}_c \tag{48}$$

where $\mathrm{TP_c}$, $\mathrm{FP_c}$, and $\mathrm{FN_c}$ denote the numbers of true positives, false positives, and false negatives for class c.

These metrics jointly reflect the overall, per-class, and region-level segmentation performance, providing a comprehensive basis for model comparison and analysis.

D. Comparison With State-of-the-Art Methods

To rigorously validate the effectiveness of VF-MET, we comprehensively compare its performance with 16 state-of-the-art semantic segmentation approaches, encompassing both unimodal and multimodal paradigms. The comparative methods include ABCNet [54], MAResU-Net [55], UNetFormer [56], TransUNet [57], CMFNet [58], vFuseNet [59], FuseNet [60], ESANet [61], SA-GATE [62], MIEFNet [63], MultiSenseSeg [64], MFNet [24], CMGFNet [65], MFTransUNet [66], FTransUNet [67], and CFCGNet [68]. Evaluations are conducted on both the Vaihingen and Potsdam datasets to ensure a comprehensive assessment.

1) Experimental Results on Vaihingen Dataset: Table I summarizes the quantitative results of VF-MET alongside 16 representative baselines on the Vaihingen dataset. VF-MET achieves the highest performance across all three key metrics: IoU (87.12%), $F1$-score (96.24%), and OA (95.88%), surpassing the second-best method, CFCGNet, by 2.23%, 4.49%, and 3.10%, respectively. This substantial performance gain is primarily attributed to the robust multimodal feature representation and the effective fusion of optical and DSM information. A detailed class-wise analysis further underscores the model's advantages. The most significant improvement is observed in the low vegetation category, where VF-MET outperforms CFCGNet by 10.78%. This is particularly notable given the high degree of visual and contextual similarity between low vegetation and tree classes, a persistent challenge for existing approaches. The TSMD in VF-MET leverages prompt-aware contextual modeling to better distinguish these ambiguous regions. In addition, in the cars category, which is characterized by small object size and high segmentation difficulty, VF-MET achieves a 6.15% improvement. The integration of MoE-MSConvLoRA empowers the encoder with enhanced multiscale sensitivity, enabling more precise segmentation of small and irregular objects. Importantly, VF-MET consistently outperforms both unimodal and multimodal competitors, demonstrating its capacity to bridge the semantic gap across modalities and to address challenges, such as class imbalance and scale variation. Fig. 5 presents visual comparisons of segmentation results in challenging scenarios. VF-MET produces sharper and more accurate boundaries, particularly in areas where tree and low vegetation overlap or are spatially adjacent. Owing to effective multimodal fusion, VF-MET also delivers superior delineation for both regular (e.g., buildings) and irregular (e.g., cars) object classes. Furthermore, the model avoids the common misclassification of nonexistent clutter regions, which remains problematic for other methods.

2) Experimental Results on Potsdam Dataset: To further assess the generalizability of VF-MET, we evaluate its performance on the Potsdam dataset using the same experimental protocols. As reported in Table II, VF-MET again achieves state-of-the-art results, with IoU, mF1, and OA of 88.31%, 97.03%, and 95.32%, respectively, outperforming

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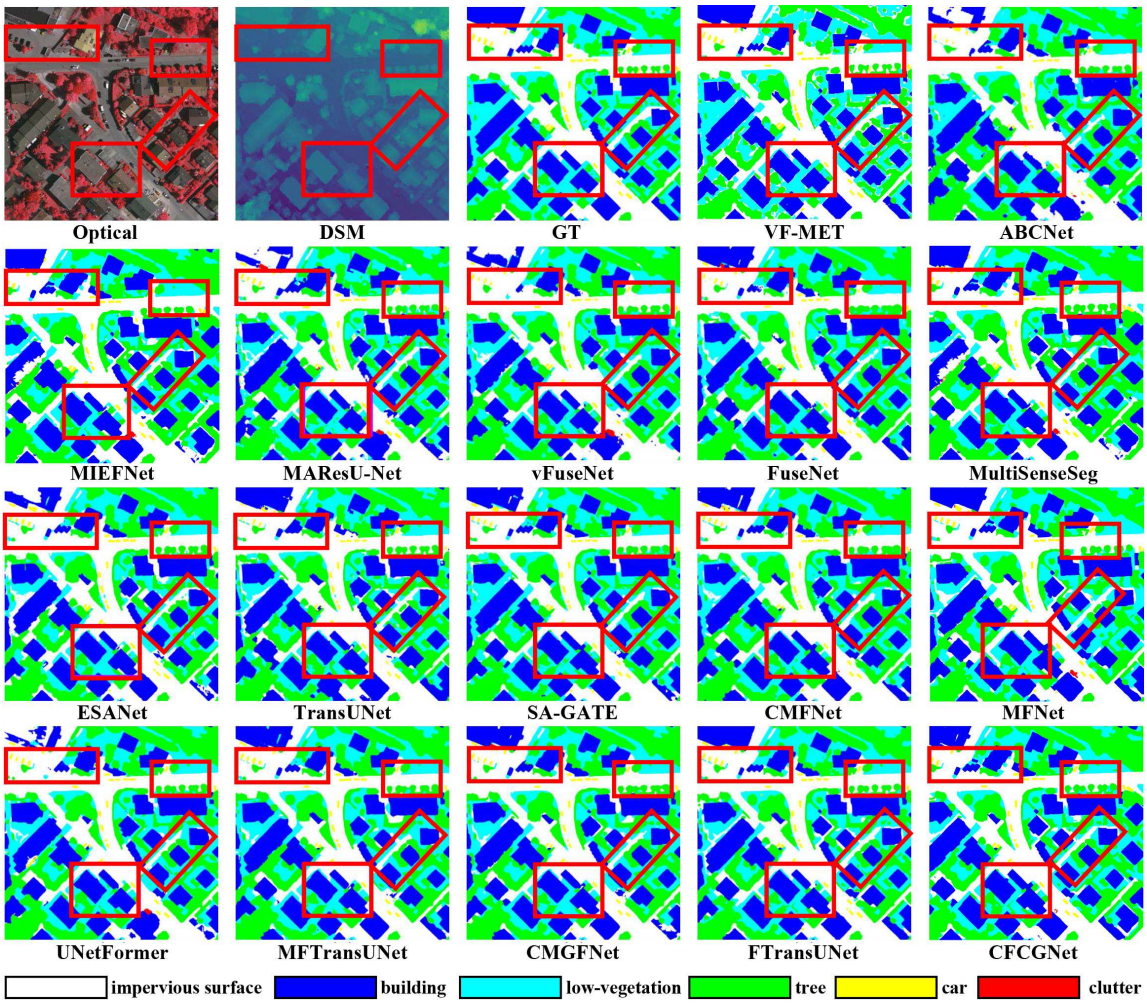


Fig. 5. Visualization segmentation results of different methods on the Vaihingen dataset.

TABLE II

EXPERIMENTAL RESULTS ON THE POTSDAM DATASET. WE PRESENT THE OA OF FIVE FOREGROUND CLASSES AND THREE OVERALL PERFORMANCE METRICS. RED VALUES INDICATE THE BEST MODEL PERFORMANCE

Type	Method	OA(%)						mF1(%)	mIoU(%)
		Bui.	Tre.	Low.	Car	Imp.	Total		
Single Modal	ABCNet	96.23	78.92	86.40	92.92	88.90	87.52	88.14	79.26
	MAResU-Net	96.82	83.97	87.70	95.88	92.19	89.82	90.86	83.61
	UNetFormer	97.69	86.47	87.93	95.91	92.27	90.65	91.71	85.05
Multi-Modal	TransUNet	96.63	82.65	89.98	93.17	91.93	90.01	90.97	83.74
	CMFNet	97.63	87.40	88.00	95.68	92.84	91.16	92.10	85.63
	vFuseNet	97.23	84.29	89.03	95.49	91.62	90.22	91.26	84.26
	FuseNet	97.48	85.14	87.31	96.10	92.64	90.58	91.60	84.86
	ESANet	97.10	85.31	87.81	94.08	92.76	89.74	91.22	84.15
	SA-GATE	96.54	81.18	85.35	96.63	90.77	87.91	90.26	82.53
	MIEFNet	97.65	86.77	89.13	95.30	91.35	90.50	91.46	84.61
	MultiSenseSeg	97.94	87.27	84.22	95.24	93.88	91.16	92.01	85.53
	MFNet	97.01	86.91	86.01	96.74	93.03	90.68	91.84	85.43
	CMGFNet	97.41	86.80	86.68	95.68	92.60	90.21	91.40	84.53
	MFTransNet	97.37	85.71	86.92	96.05	92.45	89.96	91.11	84.04
	FTransUNet	97.78	88.27	88.48	96.31	93.17	91.34	92.41	86.20
	CFCGNet	98.34	88.45	88.54	96.94	94.22	91.82	92.91	86.97
	VF-MET(Ours)	97.22	93.97	95.10	98.98	96.19	95.32	97.03	88.31

CFCGNet by 1.34%, 4.02%, and 4.50%. Notably, the Potsdam dataset introduces an additional clutter category, characterized by a small pixel proportion and highly irregular shapes. Despite this added challenge, VF-MET maintains robust

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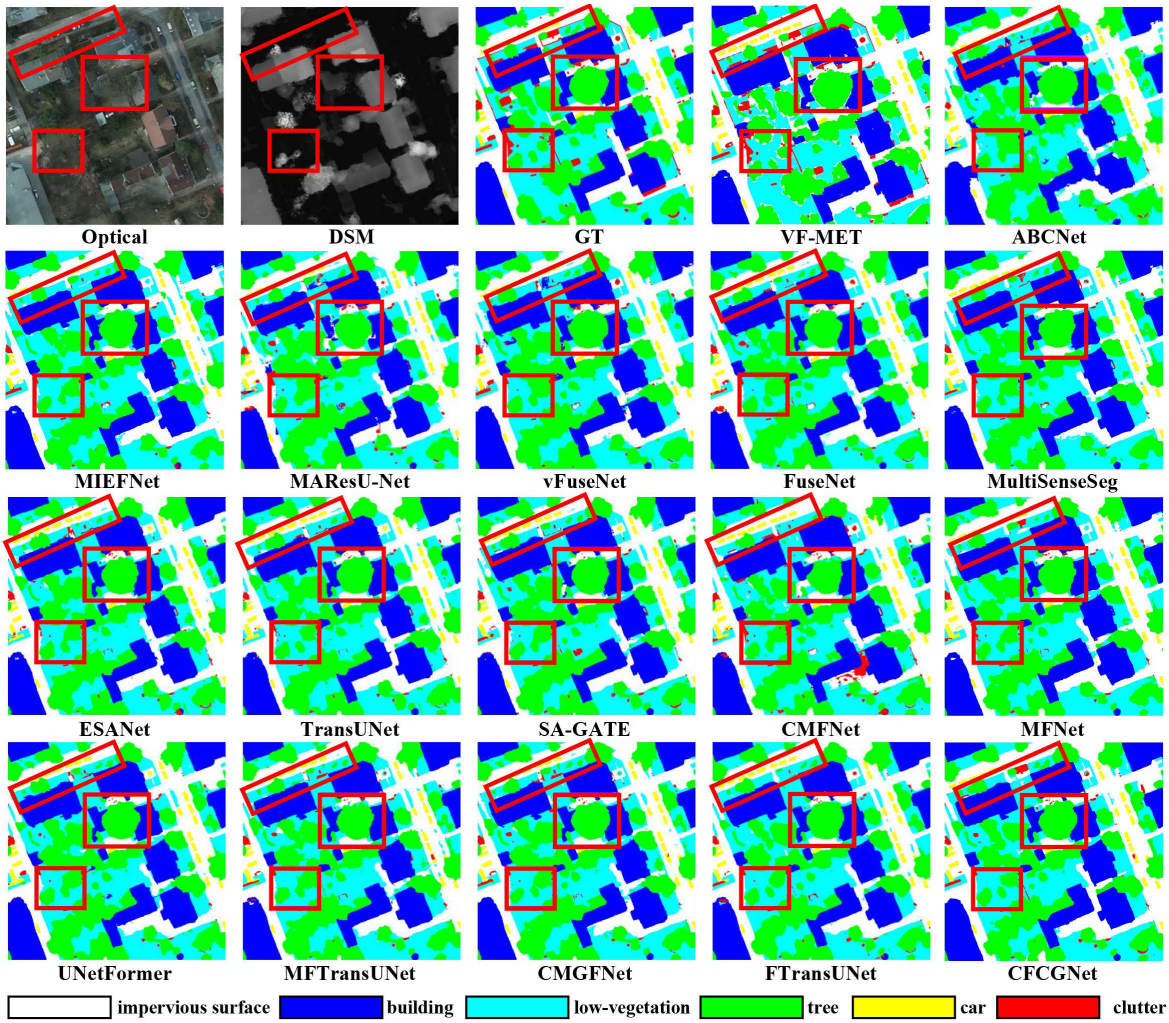


Fig. 6. Visualization segmentation results of different methods on the Potsdam dataset.

performance on clutter and all other classes, confirming its resilience to class imbalance and object diversity. VF-MET also demonstrates substantial gains in the tree and low vegetation categories, highlighting its ability to resolve fine-grained semantic ambiguities. The model excels at segmenting objects with substantial scale disparity, such as large-scale buildings and small-scale cars, further underscoring its effective multiscale representation capabilities. Fig. 6 provides visual evidence of VF-MET's superior segmentation quality on Potsdam. Across a variety of complex scenarios, VF-MET consistently produces more precise object boundaries and reduces misclassification rates for both major and minor classes. Particularly, the model accurately segments the clutter category without over-segmentation, a frequent failure mode in other methods. The collaborative box-point prompt strategy and adaptive multimodal decoder facilitate robust localization and semantic association, even in the presence of intricate object relationships and spatial arrangements. In addition, VF-MET demonstrates strong performance in categories with large intraclass variation, such as building and tree, validating its advanced contextual modeling capabilities. Collectively, these results demonstrate that VF-MET not only achieves state-of-the-art performance on both Vaihingen and Potsdam but also exhibits outstanding robustness, generalization, and adaptability across diverse object categories and scene complexities. The observed improvements can be attributed to the synergy of collaborative prompt design, expert-driven multiscale feature extraction, and a two-stage multimodal integration strategy.

E. Complexity Comparison

To assess the practical deployment potential of VF-MET, we compare its computational complexity and resource efficiency with other methods using four metrics: FLOPs, parameter count, memory footprint, and inference speed (FPS), as shown in Table III. All measurements are conducted on a single NVIDIA GeForce RTX 3090 GPU with 256×256 input. Lightweight models, such as ABCNet, MFTransUNet, and FuseNet achieve the highest inference speeds and lowest resource consumption. For example, ABCNet reaches 15.87 FPS with just 13.39 M parameters. In contrast, VF-MET achieves a competitive 8.72 FPS, balancing efficiency and accuracy among multimodal models. VF-MET's moderate computational cost mainly arises from its two-stage cross-modal transformer design and prompt-based fusion, which introduce some overhead compared to simpler architectures.

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TABLE III
COMPUTATIONAL COMPLEXITY ANALYSIS MEASURED BY A 256 × 256 IMAGE ON A SINGLE NVIDIA GEFORCE RTX 3090 GPU. RED VALUES INDICATE THE BEST MODEL PERFORMANCE

Type	Method	FLOPs (G)	Param (M)	Memory (MB)	Speed (FPS)	mIoU (%)
Single-Modal	ABCNet	3.9	13.39	1598	15.87	79.26
	MAResU-Net	8.79	26.27	1908	10.62	83.61
	UNetFormer	6.04	24.20	1980	13.89	85.05
Multi-Modal	vFuseNet	60.36	44.17	2618	16.93	83.74
	FuseNet	58.37	42.08	2284	18.92	85.63
	ESANet	7.73	34.03	1914	10.42	84.26
	MIEFNet	35.33	127.54	3276	7.22	84.61
	MultiSenseSeg	67.21	186.77	3921	4.81	85.53
	MFNet	48.21	105.06	6854	18.52	85.43
	TransUNet	32.27	93.23	3028	10.81	84.86
	SA-GATE	41.28	110.85	3174	10.00	84.15
	CMFNet	78.25	123.63	4058	8.62	82.53
	MFTransUNet	8.44	43.77	1549	14.88	84.53
	CMGFNet	19.51	64.20	2463	11.61	84.04
	FTransUNet	45.21	160.88	3463	9.74	86.20
	CFCGNet	41.43	114.54	3543	11.87	86.97
	VF-MET (Ours)	55.61	91.26	2833	8.72	88.31

TABLE IV
ABLATION STUDY OF THE PROPOSED MAJOR MODULES ON THE POTSDAM DATASET

Model	DSM	MSConvLoRA	TSMD	OA	mF1	mIoU
Baseline	×	×	×	89.93	90.48	84.70
Baseline + DSM (Concat + Conv Fusion)	✓	×	×	91.66	92.95	85.52
Baseline + MSConvLoRA + DSM (Concat + Conv Fusion)	✓	✓	×	93.07	95.22	86.86
Baseline + MSConvLoRA + DSM + TSMD	✓	✓	✓	95.32	97.03	88.31

However, thanks to LoRA-based fine-tuning and weight sharing, VF-MET significantly reduces parameter count (41.43 M) and memory usage (1549 MB), outperforming other high-accuracy models, such as CFCGNet and FTransUNet in resource efficiency while maintaining state-of-the-art accuracy (88.31% mIoU). Overall, VF-MET achieves an excellent tradeoff between segmentation performance and computational efficiency, making it well-suited for real-world applications.

F. Ablation Studies

To thoroughly investigate the contributions of each component in our framework, we conduct extensive ablation studies on the Potsdam and Vaihingen datasets.

1) Ablation of Major Modules: We first evaluate the effectiveness of the key modules in VF-MET on the Potsdam dataset. As shown in Table IV, the baseline configuration corresponds to the original SAM applied solely to optical images, resulting in an mIoU of 84.70%. Incorporating the DSM branch with feature concatenation and convolutional fusion increases mIoU by 0.82%, highlighting the benefit of multimodal 3-D information. Introducing the MoE-MSConvLoRA fine-tuning strategy further boosts mIoU by 1.34%, underscoring the value of domain-adaptive, multiscale parameter adjustment for handling complex urban objects. Finally, equipping the decoder with the TSMD provides an additional 1.45% improvement, confirming the efficacy of our two-stage cross-modal fusion design. The visual effects of these module additions are illustrated in Fig. 7. The upper part of the figure demonstrates, step by step, how the inclusion of MoE-MSConvLoRA and the subsequent PF-Fusion and FF-Fusion modules progressively enhance feature representations. Notably, the final output with all modules exhibits clearer object boundaries and more accurate delineation of complex urban structures, closely matching the ground truth.We further perform "reverse ablation" by removing each module in turn (Table V). Excluding the DSM branch leads to the most significant drop in performance, revealing the critical role of 3-D information. Removing the TSMD brings a decrease of 2.70% in mIoU, indicating that simple feature concatenation is insufficient to fully exploit multimodal cues. The fine-tuning strategy also offers a substantial benefit, though its impact is secondary to that of the advanced decoder.

2) Ablation of Fine-Tuning Strategies: To further validate the superiority of our proposed fine-tuning approach, we compare MoE-MSConvLoRA to other mainstream strategies on the Vaihingen dataset (Table VI). The results show that our approach yields the most precise and complete segmentation, particularly for small and irregular objects, whereas alternative strategies tend to produce incomplete or noisy segmentation maps. The adapter-based method achieves the lowest performance, even falling below that of "no fine-tuning," likely due to the absence of residual connections that impair the

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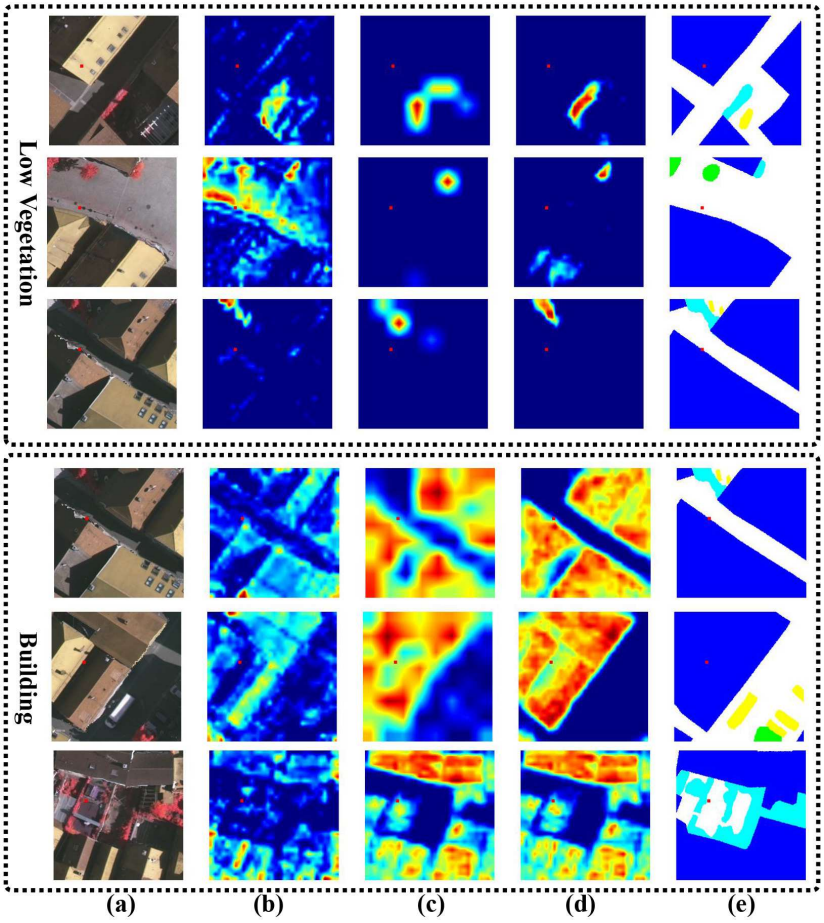


Fig. 7. Visual of major modules. (a) Optical image. (b) Baseline with MSConvLoRA. (c) Baseline with PF-Fusion. (d) Baseline with FF-Fusion. (e) GT.

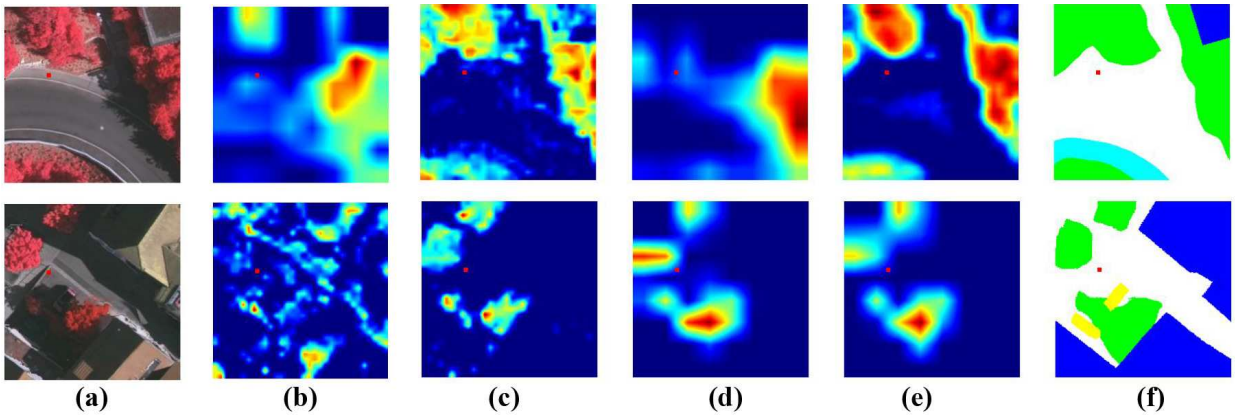


Fig. 8. Visual comparison of fine-tuning strategies. (a) Optical image. (b) Adapter. (c) LoRA. (d) ConvLoRA. (e) MoE-MSConvLoRA (ours). (f) GT.

TABLE V
REVERSE ABLATION STUDY OF THE PROPOSED MAJOR MODULES ON THE POTSDAM DATASET

Model	TSMD	MSConvLoRA	DSM	OA	mF1	mIoU
VF-MET + w/o TSMD + (Concat + Conv Fusion)	×	✓	✓	93.07	95.22	86.86
VF-MET + w/o MSConvLoRA	✓	×	✓	93.77	96.92	87.10
VF-MET + w/o DSM	×	✓	×	91.19	93.04	85.61
VF-MET	✓	✓	✓	95.32	97.03	88.31

transfer of pretrained knowledge. ConvLoRA outperforms standard LoRA by better capturing irregular object features, especially when combined with our hybrid expert model. Nevertheless, only MoE-MSConvLoRA achieves the highest

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TABLE VI
ABLATION STUDY OF DIFFERENT FINE-TUNING STRATEGIES ON THE VAIHINGEN DATASET

Model	TSMD	MSConvLoRA	DSM	OA	mF1	mIoU
VF-MET + w/o MSConvLoRA + LoRA	✓	✓	✓	94.39	95.07	87.22
VF-MET + w/o MSConvLoRA + Adapter	✓	✓	✓	92.04	94.36	86.03
VF-MET + w/o MSConvLoRA + ConvLoRA	✓	✓	✓	94.13	95.91	87.78
VF-MET	✓	✓	✓	95.32	97.03	88.31

TABLE VII
REVERSE ABLATION STUDY OF DIFFERENT PROMPT LEARNING STRATEGIES ON THE VAIHINGEN DATASET

Model	TSMD	MSConvLoRA	DSM	OA	mF1	mIoU
VF-MET + w/o Our Prompt + Box Prompt	✓	✓	✓	83.88	85.13	79.81
VF-MET + w/o Our Prompt + Point Prompt	✓	✓	✓	92.61	94.11	87.07
VF-MET (Box and Point Prompt)	✓	✓	✓	95.32	97.03	88.31

mIoU, demonstrating its advantage in learning domain-specific knowledge.As illustrated in Fig. 8, the visual comparison further highlights the strengths and weaknesses of each fine-tuning strategy. The adapter and LoRA-based approaches often yield incomplete or noisy segmentation, especially for small-scale or irregularly shaped objects. ConvLoRA achieves better localization, but only the proposed MoE-MSConvLoRA produces segmentation maps that closely align with the ground truth, effectively capturing fine details and object boundaries. This visual evidence is consistent with the quantitative improvements observed in Table VI, and further demonstrates the superiority and robustness of our expert-driven fine-tuning strategy in complex multimodal scenarios.

3) Ablation of Prompt Learning Strategies: Prompt learning plays a crucial role in interactive segmentation for multimodal remote sensing. To thoroughly evaluate the effectiveness of different prompt types, we conducted comprehensive ablation experiments, as shown in Table VII. When only box prompts are used, the model achieves an mIoU of 79.81%, which is significantly lower than the other configurations. This indicates that box prompts alone provide insufficient spatial cues for accurate segmentation, likely due to high object density and weak spatial correlation in remote sensing imagery, leading to frequent overlaps and ambiguities. Using only point prompts increases the mIoU to 87.07%, demonstrating that point prompts are more effective at capturing localized object features and adapting to ambiguous or small object boundaries, especially when combined with the elevation information from the DSM modality. However, the best performance is achieved when both box and point prompts are integrated; our CBPM achieves an mIoU of 88.31%, with corresponding improvements in OA and mF1. This approach outperforms the box-only and point-only strategies by 8.5% and 1.24% in mIoU, respectively. These quantitative results demonstrate that the collaborative prompt strategy leverages the complementary strengths of both prompt types; box prompts provide coarse spatial localization, while point prompts offer fine-grained contextual refinement.This synergy enables the model to better distinguish densely packed or overlapping objects and improves overall segmentation robustness in complex multimodal remote sensing scenarios. In addition, Fig. 9 provides qualitative comparisons of segmentation activation maps under different prompt strategies. The collaborative box-point approach generates more precise and complete object boundaries than either single-type prompt, further validating the effectiveness of our mechanism.

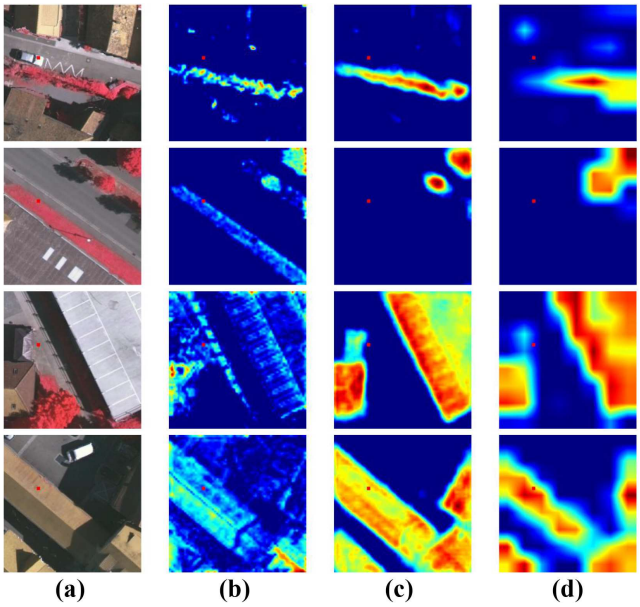


Fig. 9. Comparison of segmentation activation maps with different prompt strategies. (a) Optical image. (b) Only box prompts. (c) Our box-point prompts. (d) Only point prompts.

G. Cross-Region Generalization Analysis

To further assess the cross-region generalization capability of VF-MET, we conduct experiments on the Potsdam dataset, as summarized in Table VIII. Specifically, six regions, [2_13, 3_14, 4_15, 5_13, 6_14, and 7_13], are selected and excluded from both the training and validation sets. The models are evaluated exclusively on these previously unseen

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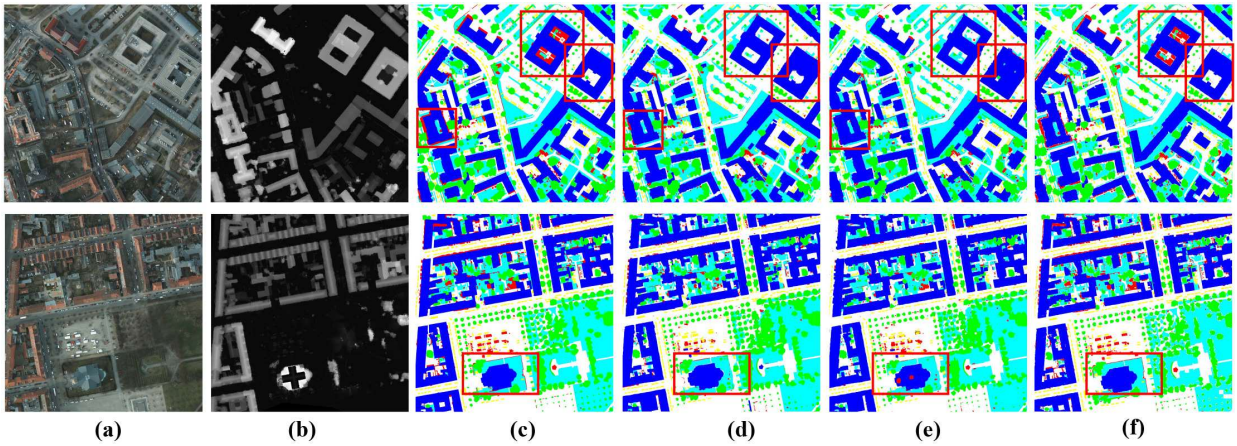


Fig. 10. Visualization of cross-region generalization analysis. (a) Optical image. (b) DSM. (c) GT. (d) CFCGNet. (e) FTransUNet. (f) VF-MET.

TABLE VIII
CROSS-REGION GENERALIZATION ANALYSIS ON THE POTSDAM DATASET (UNSEEN REGIONS). WE REPORT THE OA OF FIVE FOREGROUND CLASSES AND THREE OVERALL METRICS

Method	Bui.	Tre.	Low.	Car	Imp.	Total	mF1	mIoU
FTransUNet	97.38	85.78	91.39	95.15	94.22	91.92	92.64	86.52
CFCGNet	98.25	89.55	89.68	97.32	94.50	92.69	93.13	87.21
VF-MET	97.50	94.89	95.73	98.69	96.94	96.98	98.39	89.76

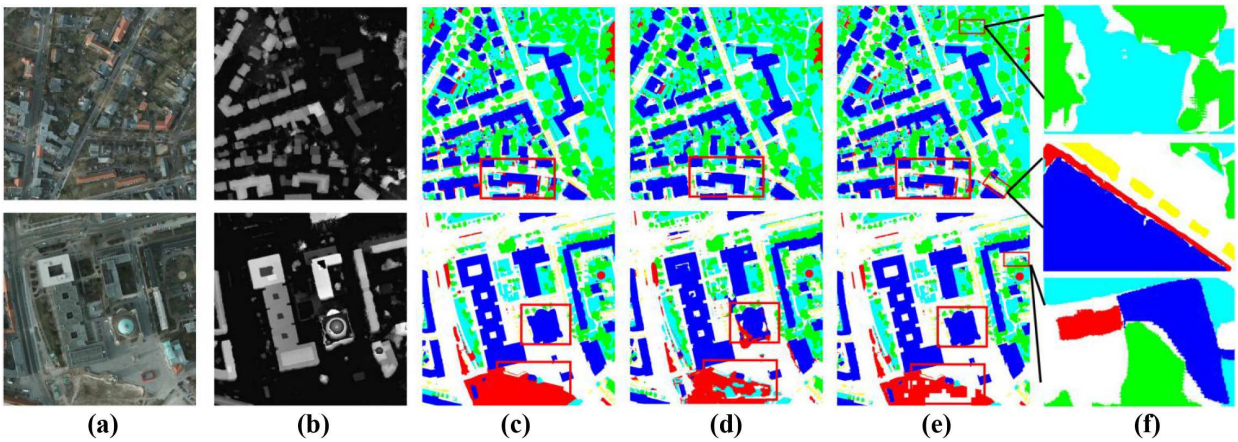


Fig. 11. Visualization of failure case analysis. (a) Optical image. (b) DSM. (c) GT. (d) CFCGNet. (e) VF-MET. (f) Enlarged region.

regions to rigorously assess their segmentation performance in new geographic areas. Our approach is compared against two strong multimodal baselines, CFCGNet and FTransUNet. VF-MET achieves an mIoU of 89.76%, surpassing CFCGNet by 2.55%. This substantial improvement underscores the superior cross-region generalization ability of VF-MET, which can be largely attributed to the intrinsic zero-shot learning strengths of the SAM backbone. In addition, the multimodal extensions and advanced fine-tuning strategies in VF-MET further enhance its capacity to transfer knowledge to novel geographic regions within the same dataset. Qualitative results are presented in Fig. 10. Across all categories, VF-MET consistently produces the most accurate segmentation, with particularly notable improvements in challenging classes, such as clutter and building. These classes are characterized by irregular shapes and significant scale variations, which typically pose difficulties for conventional models. The superior performance of VF-MET in these cases can be ascribed to the advanced fine-tuning strategy, which effectively models irregular and multiscale structures, as well as the decoder's ability to capture contextual relationships and distinctions between different land cover types. Overall, both quantitative and qualitative results demonstrate that VF-MET possesses robust cross-region generalization capability in remote sensing scenarios, and is well-suited for deployment in previously unseen geographic regions from the same modality and dataset.

H. Failure Case Analysis

While VF-MET demonstrates substantial improvements over previous methods, especially in accurately segmenting

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challenging categories, such as clutter, where models like CFCGNet have exhibited significant deficiencies, certain limitations remain. The enhanced performance in the clutter category can be attributed to the MoE-MSConvLoRA module, which effectively addresses the irregular shapes and pronounced scale variations inherent to these scenes, resulting in more precise and robust segmentation. However, as illustrated in Fig. 11, our approach still exhibits noticeable boundary artifacts. Although segmentation results are generally accurate at the object level, they consistently present jagged and irregular boundaries across all categories. This phenomenon, also observed in our previous work [32], has only recently been recognized as a critical issue warranting deeper investigation. The primary cause of this limitation lies in the fundamental characteristics of the underlying visual foundation model. SAM, built entirely upon the Transformer architecture, excels at global feature modeling but is less effective at capturing fine-grained spatial details and precise object contours. Consequently, the model's inherent capacity for boundary delineation is limited. Addressing the challenge of boundary refinement in multimodal remote sensing segmentation will thus be a key direction for future research. Potential solutions may include the integration of dedicated boundary-aware modules or hybrid designs that combine the global modeling strengths of Transformers with the local detail-preserving capabilities of convolutional neural networks.

V. Conclusion

In this article, we presented VF-MET, a novel VF-MET framework for multimodal remote sensing semantic segmentation. VF-MET systematically tackles key challenges, such as modality gaps, scale variation, and irregular object boundaries in multimodal scenarios. Leveraging an AMET strategy based on MoE-MSConvLoRA, our approach enables efficient and robust extraction of heterogeneous features across both modalities and scales. The CBPM further improves segmentation accuracy, particularly for dense and weakly correlated objects, while the TSMD facilitates effective multimodal fusion and contextual integration. Extensive experiments on benchmark datasets demonstrate that VF-MET achieves state-of-the-art performance, surpassing leading unimodal and multimodal methods in accuracy, robustness, and generalization. Ablation studies validate the contribution of each core module, and cross-region evaluations highlight strong zero-shot transfer capabilities. Future work will focus on enhancing boundary refinement and extending the framework to broader multimodal sensing applications and large-scale Earth observation tasks.

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